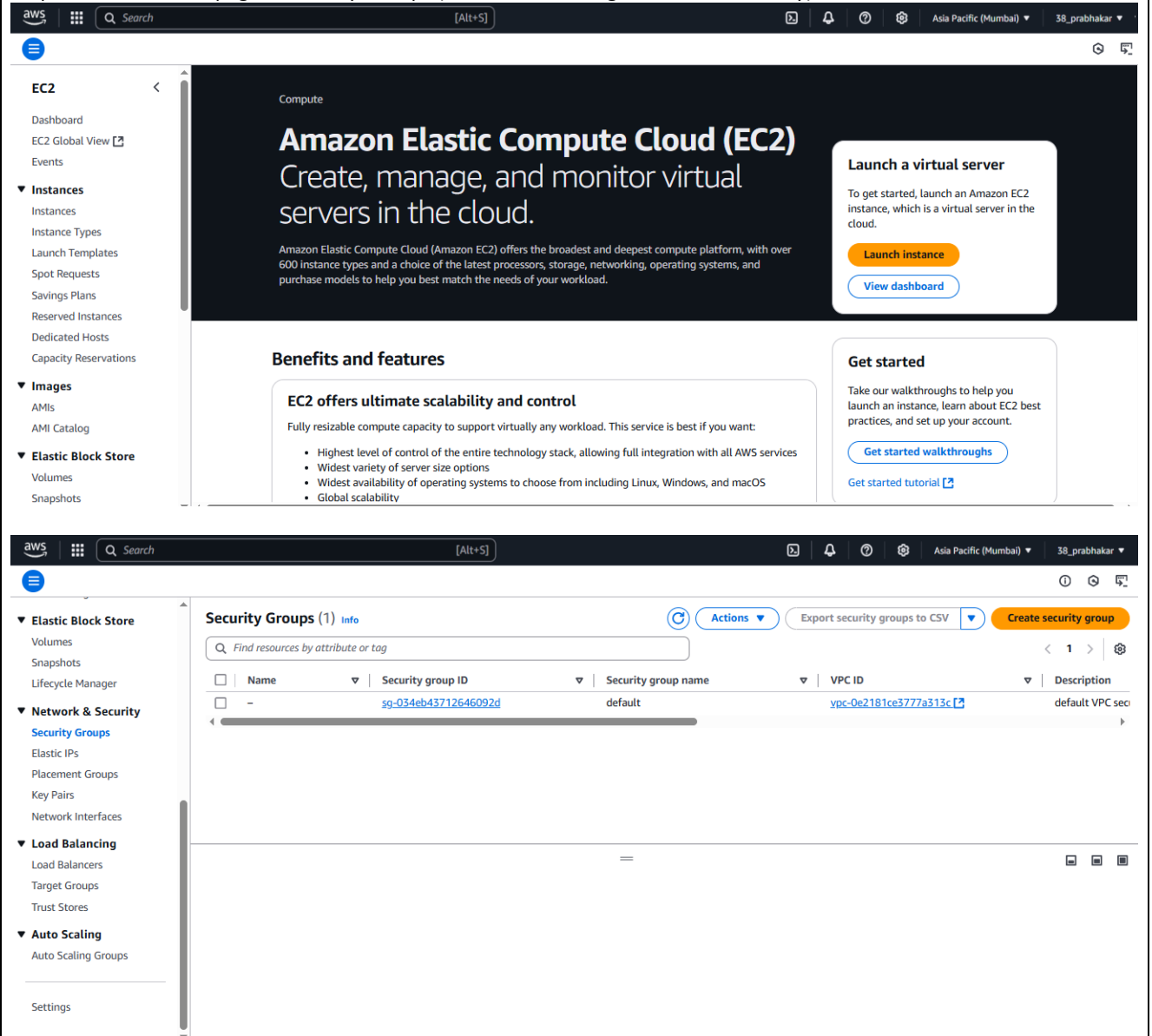


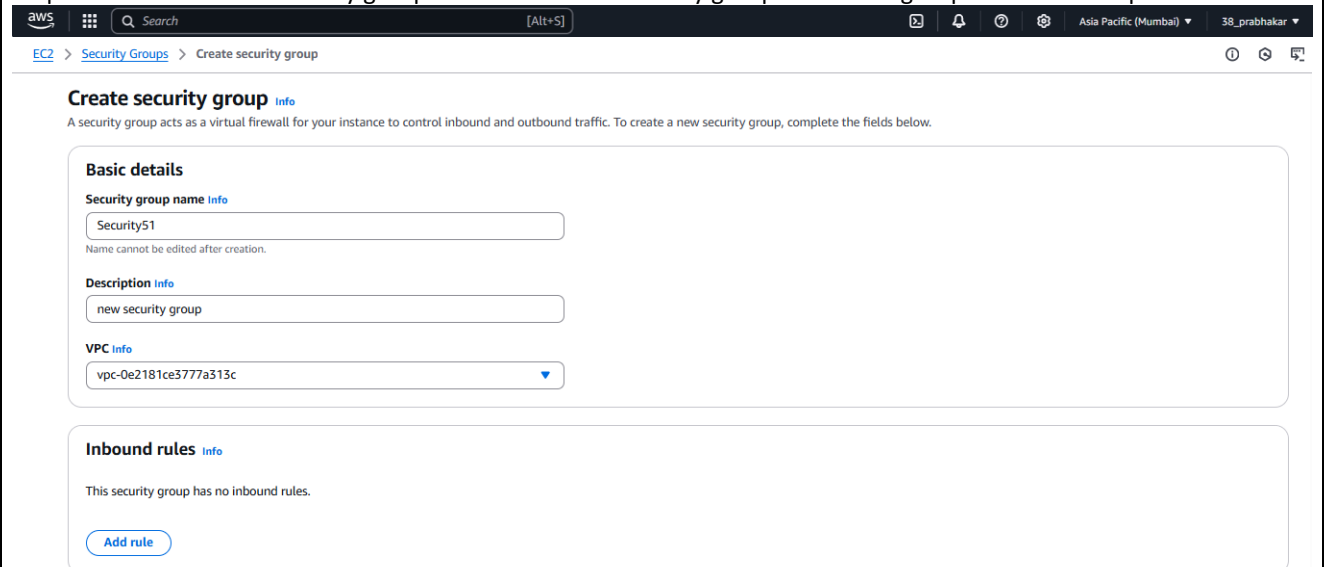
Assignment : 11

Build scaling plans in AWS that balance the load on different EC2 instances.

Step 1: Go to the EC2 page > Security Groups (under the heading Network & Security).



Step 2: Click on “Create security group” to create a new security group. Name the group and add description.



Step 3: Add the below 4 inbound rules (SSH, HTTP, HTTPS, Custom TCP) and select Anywhere IPv4 in the Source. Add port range 4000 in the Custom TCP type. Afterwards click on Create Security Group.

aws Search [Alt+S] Asia Pacific (Mumbai) 38_prabhakar

EC2 > Security Groups > Create security group

Inbound rules

Type	Protocol	Port range	Source	Description - optional	Actions
SSH	TCP	22	Anyw... 0.0.0.0/0		Delete
HTTP	TCP	80	Anyw... 0.0.0.0/0		Delete
HTTPS	TCP	443	Anyw... 0.0.0.0/0		Delete
Custom TCP	TCP	4000	Anyw... 0.0.0.0/0		Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

aws Search [Alt+S] Asia Pacific (Mumbai) 38_prabhakar

EC2 > Security Groups > sg-0026f27c5ffcd4d49 - Security51

EC2 Dashboard EC2 Global View Events

Instances Instances Instance Types Launch Templates Spot Requests Savings Plans Reserved Instances Dedicated Hosts Capacity Reservations

Images AMIs

Security group (sg-0026f27c5ffcd4d49 | Security51) was created successfully

sg-0026f27c5ffcd4d49 - Security51

Details

Security group name Security51	Security group ID sg-0026f27c5ffcd4d49	Description new security group	VPC ID vpc-0e2181ce3777a313c
Owner 586794457897	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

Inbound rules (4) Manage tags Edit inbound rules

Step 4: From the lefthand side menu bar, go to Launch Templates and click on "Launch Template".

aws Search [Alt+S] Asia Pacific (Mumbai) 38_prabhakar

EC2 > Security Groups > sg-0026f27c5ffcd4d49 - Security51

EC2 Dashboard EC2 Global View Events

Instances Instances Instance Types Launch Templates Spot Requests

Launch Templates

You do not have any Launch Templates in this region

Step 5: Name the Template and set the version as "v0", also check the Auto scaling guidance box.

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EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required
Template51
Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '!', '!', '@'.

Template version description
v0
Max 255 chars

Auto Scaling guidance
Select this if you intend to use this template with EC2 Auto Scaling
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags
Source template

Summary

- Software image (AMI)
- Virtual server type (instance type)
- Firewall (security group)
- Storage (volumes)

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Step 6: Scroll down to Application and OS > Quick Start > Ubuntu.

Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents | **Quick Start**

Amazon Linux | macOS | **Ubuntu** | Windows | Red Hat | SUSE Linux | Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0e35ddab05955cf57 (64-bit (x86)) / ami-0429d68a1cd41ca80 (64-bit (ARM))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0e35ddab05955cf57

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Create launch template](#)

Step 7: In the instance type, select t2.micro as it is the Free tier eligible instance.

Instance type [Info](#) | [Get advice](#) [Advanced](#)

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour
On-Demand SUSE base pricing: 0.0124 USD per Hour

Free tier eligible

☒ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Don't include in launch template [Create new key pair](#)

Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template [Create new subnet](#)

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0e35ddab05955cf57

Virtual server type (instance type)
t2.micro

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Create launch template](#)

Step 8: Create a new key pair for login.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
key-51
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

Step 9: Network Settings > Firewall (security groups) > Select Existing groups. Now select the above made security group.

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EC2 > Launch templates > Create launch template

Network settings

Subnet | Info

Don't include in launch template

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group Create security group

Security groups | Info

Select security groups

Specify a custom value...

default sg-034eb43712646092d

Security51 sg-0026f27c5ffcd4d49

Compare security group rules

Hide details

Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3)

AMI Volumes are not included in the template unless modified

Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955c5f57

Virtual server type (instance type)

t2.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Create launch template

Step 10: In the Advanced details, write the below mentioned code,

```
#!/bin/bash
apt-get update
apt-get upgrade
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/Albega34/Psg_repo.git
cd Psg_repo
npm install
node index.js
```

And the click on "Create Launch Template".

aws Search [Alt+S] United States (N. Virginia) 38_prabhakar

EC2 > Instances > Launch an instance

Select

User data - optional | Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt-get update
apt-get upgrade
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/Albega34/Psg_repo.git
cd Psg_repo
npm install
node index.js
```

User data has already been base64 encoded

Summary

Number of instances | Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...read more

ami-0e449927258d45bc4

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance

Preview code

aws Search [Alt+S] Asia Pacific (Mumbai) 38_prabhakar

EC2 > Launch templates > Create launch template

Success

Successfully created [Template51\(lt-0edc13f38e715efa6\)](#).

Actions log

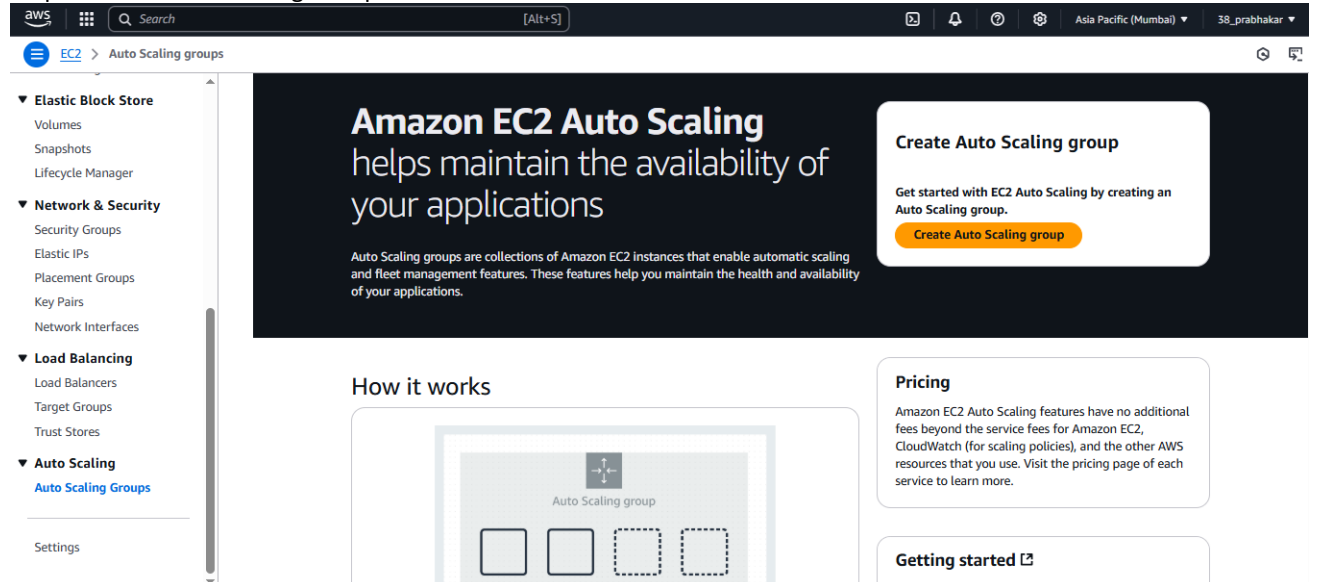
Next Steps

Launch an instance

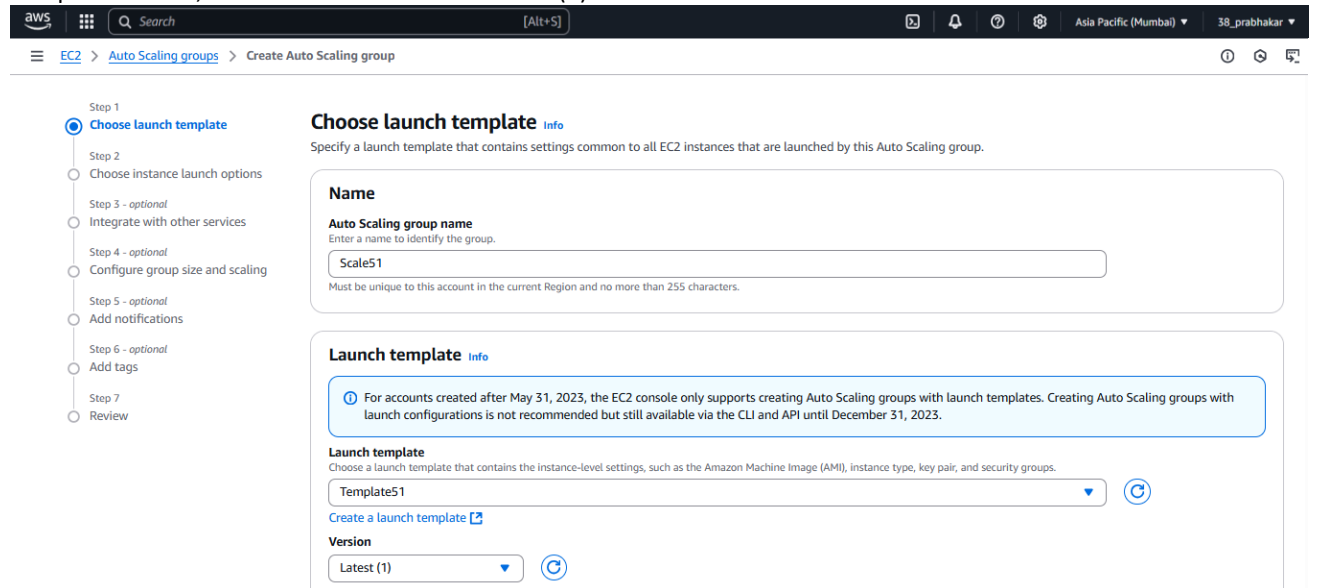
With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.

[Launch instance from this template](#)

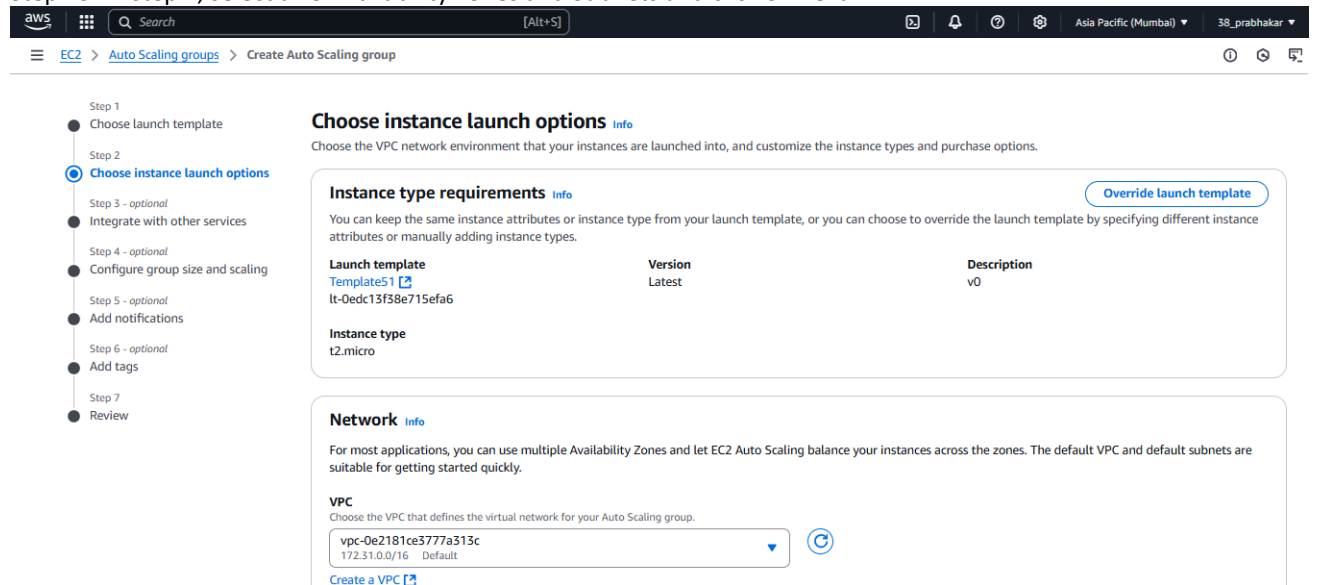
Step 11: Go to Auto Scaling Groups in the lefthand side menu.



Step 12: Click on “Create Auto Scaling Group”, name the group, select the above made template in the Launch Template section, and select the version as Latest(1).



Step 13: In step 2, select all 3 Availability Zones and Subnets and click on next.



aws

Search

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Asia Pacific (Mumbai)

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 7
Review

Network info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0e2181ce3777a313c
172.31.0.0/16 Default

Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets

ap-south-1a | subnet-02fbd93920842d759
172.31.32.0/20 Default

ap-south-1b | subnet-0e06b1c09b402835f
172.31.0.0/20 Default

ap-south-1c | subnet-0542c368c04490950
172.31.16.0/20 Default

Create a subnet

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

Balanced only

Step 14: In step 3, select the below shown options and Create a new target group in Default Routing and click on Next.

aws

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Integrate with other services - optional info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ Attach to an existing load balancer
Choose from your existing load balancers.

☒ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to a new load balancer
Define a new load balancer to create for attachment to this Auto Scaling group.

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the [Load Balancing console](#).

☐ Application Load Balancer
HTTP, HTTPS

☒ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.
Scale51-1

aws

Search

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Load balancer name
Name cannot be changed after the load balancer is created.
Scale51-1

Load balancer scheme
Scheme cannot be changed after the load balancer is created.

☐ Internal

☒ Internet-facing

Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

VPC
vpc-0e2181ce3777a313c

Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

☒ ap-south-1b
subnet-0e06b1c09b402835f

☒ ap-south-1c
subnet-0542c368c04490950

☒ ap-south-1a
subnet-02fbd93920842d759

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol	Port	Default routing (forward to)
----------	------	------------------------------

VPC
vpc-0e2181ce3777a313c

Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

- ☒ ap-south-1b: subnet-0e06b1c09b402835f
- ☒ ap-south-1c: subnet-0542c368c04490950
- ☒ ap-south-1a: subnet-02fbd93920842d759

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol: TCP | **Port**: 80 | **Default routing (forward to)**: Create a target group

New target group name
An instance target group with default settings will be created.
Scale51-1

Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.
[Add tag](#) (50 remaining)

Step 15: In step 4, Select Desired capacity as 2, Min desired capacity as 2 and Max desired capacity as 3.

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)

Desired capacity
Specify your group size.
2

Scaling
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity: 2 (Equal or less than desired capacity) | **Max desired capacity**: 3 (Equal or greater than desired capacity)

Step 16: In Automatic Scaling, Choose Target tracking scaling policy, set the policy name, Target value as 32 and Instance Warmup as 300 seconds and click on Next.

Automatic scaling - optional
Choose whether to use a target tracking policy

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name
Target Tracking Policy

Metric type
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization

Target value
32

Instance warmup
300 seconds

☐ Disable scale in to create only a scale-out policy

Instance maintenance policy

Step 17: In step 5 & 6, click on Next. In step 7, after reviewing the whole Auto Scaling Group, click on Create Auto Scaling Group.

[illegible]




[Alt+S]




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[EC2](#) > Auto Scaling groups





Auto Scaling groups (1) [Info](#)


[Launch configurations](#)
[Launch templates](#)
[Actions ▾](#)
[Create Auto Scaling group](#)




☐	Name ▾	Launch template/configuration ▾	Instances ▾	Status ▾	Desired capacity ▾	Min ▾	Max ▾	Availability Zones ▾
☐	Scale51	Template51 Version Latest	2	-	2	2	3	ap-south-1c, ap-south...

Step 18: Go to Instances.

aws

Search

[Alt+S]

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EC2

Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Instances (2) info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☐		i-0da4332b356f62c28	Running	t2.micro	Initializing	View alarms +	ap-south-1a	ec2-13-2
☐		i-0d6083ca069a8335c	Running	t2.micro	Initializing	View alarms +	ap-south-1b	ec2-3-7-

Step 19: Click on the first instance and copy its IPv4 address and run it in a separate incognito window to check if it's functional or not.

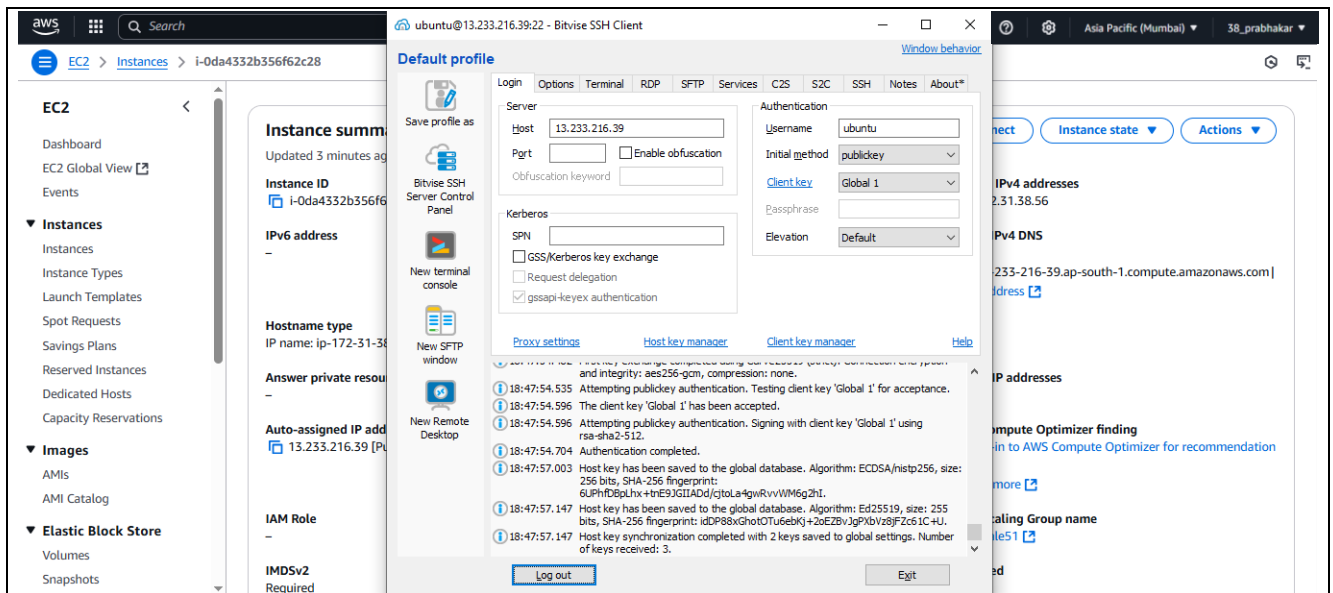
The screenshot shows the AWS Management Console interface for an EC2 instance. The left sidebar contains navigation links for EC2, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, and Snapshots. The main content area displays the 'Instance summary for i-0da4332b356f62c28'. Key details include: Instance ID (i-0da4332b356f62c28), IPv6 address (none), Hostname type (IP name: ip-172-31-38-56.ap-south-1.compute.internal), Answer private resource DNS name (none), Auto-assigned IP address (13.233.216.39 [Public IP]), IAM Role (none), IMDSv2 (Required), Public IPv4 address (13.233.216.39 | open address), Instance state (Running), Private IP DNS name (IPV4 only) (ip-172-31-38-56.ap-south-1.compute.internal), Instance type (t2.micro), VPC ID (vpc-0e2181ce3777a313c), Subnet ID (subnet-02fbd9392084d759), Instance ARN (arn:aws:ec2:ap-south-1:586794457897:instance/i-0da4332b356f62c28), Private IPv4 addresses (172.31.38.56), Public IPv4 DNS (ec2-13-233-216-39.ap-south-1.compute.amazonaws.com | open address), Elastic IP addresses (none), AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations. | Learn more), Auto Scaling Group name (Scale51), and Managed (false).

Below the console, a web browser window is open to the URL 13.233.216.39. The browser displays a 'Welcome to nginx!' message, indicating that the nginx web server is successfully installed and working. The message also provides links for online documentation and support, and a thank you note for using nginx.

Step 20: Copy the same IPv4 address in Bitwise SSH Client, import the newly made key pair and Log in.

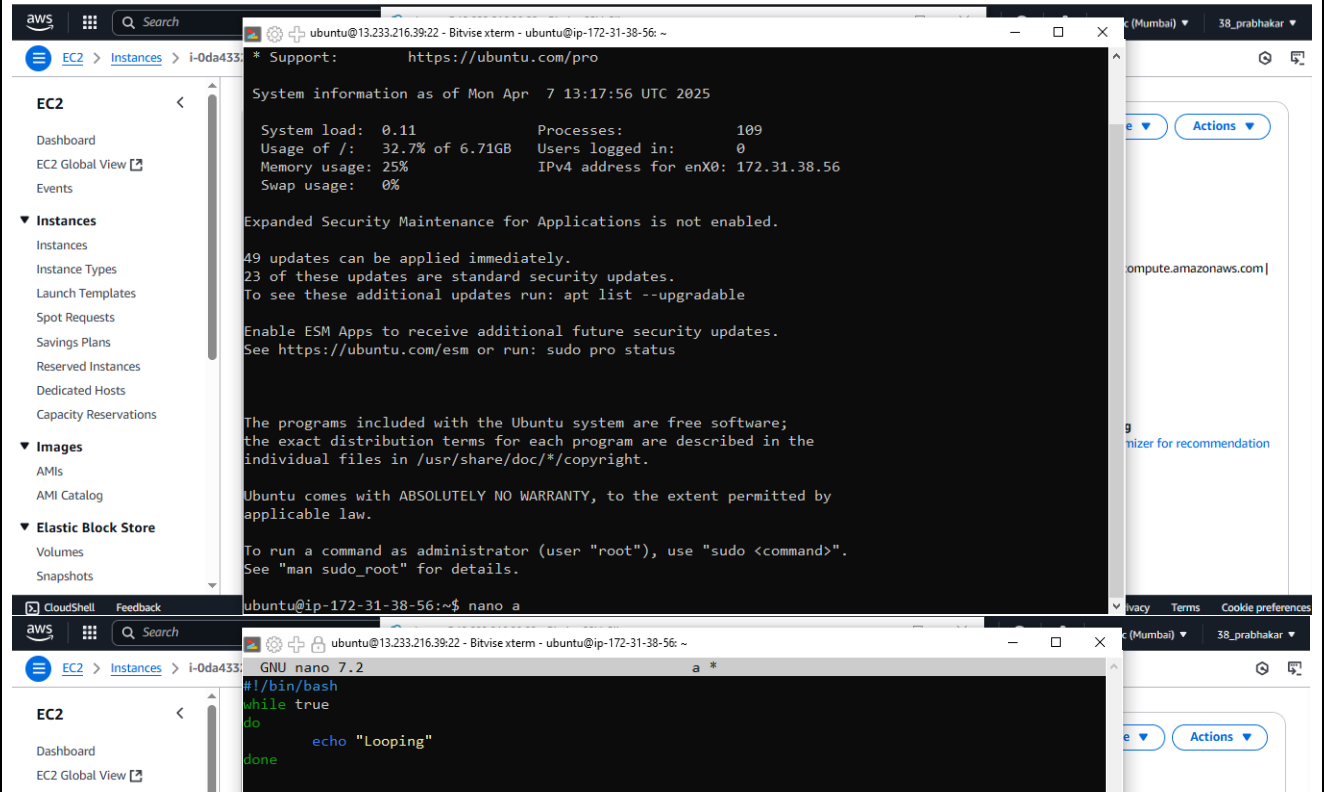
The screenshot shows the AWS Management Console interface for an EC2 instance, the Bitwise SSH Client window, and a terminal window. The AWS console shows the same instance details as in Step 19. The Bitwise SSH Client window is open, showing the 'Default profile' tab. The 'Server' field is set to 13.233.216.39. The 'Authentication' tab is selected, showing the 'Username' as 'ubuntu' and the 'Initial method' as 'publickey'. The 'Client key' is set to 'Global 1'. The 'Elevation' is set to 'Default'. The 'Log in' button is visible. The terminal window is open, showing the output of the 'ssh' command, which includes a warning about a known host and a message indicating that the connection is successful.

Step 21: Open a new Terminal.

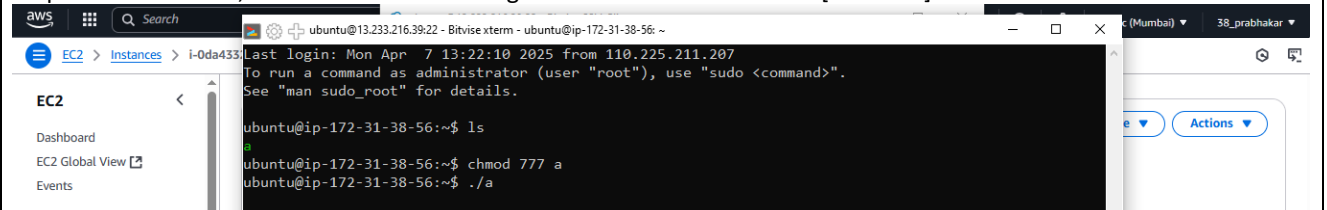


Step 22: Create a new file using “nano” command and write the following block of code,

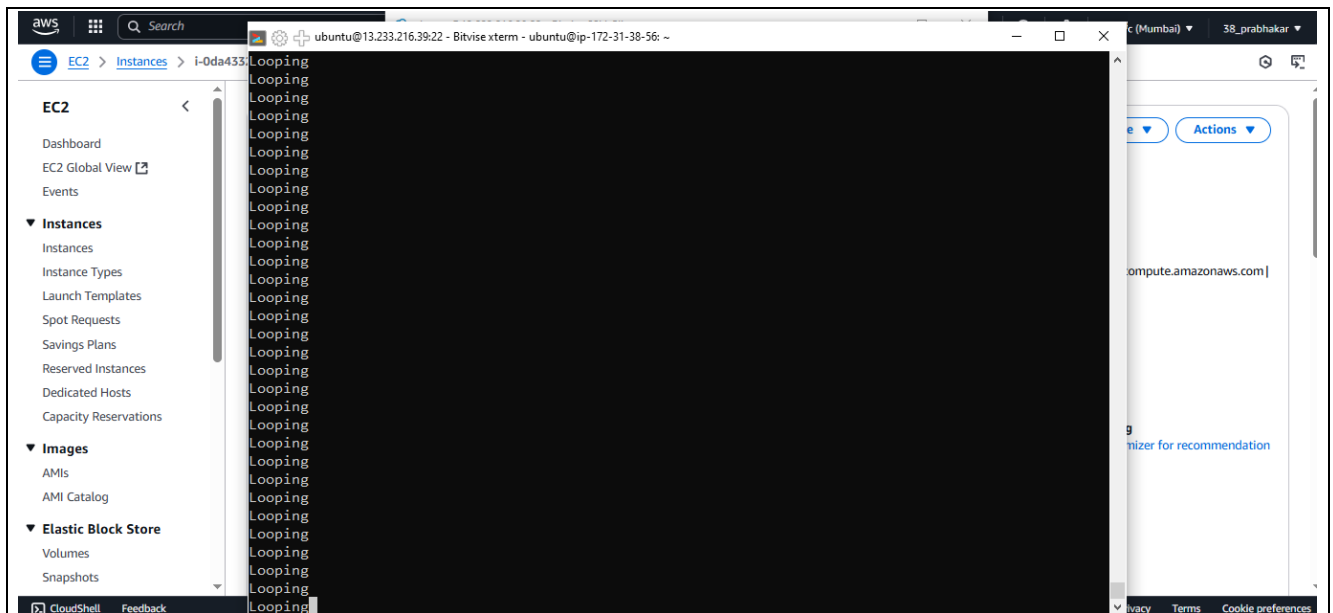
```
#!/bin/bash
while true
do
    echo "Looping"
done
```



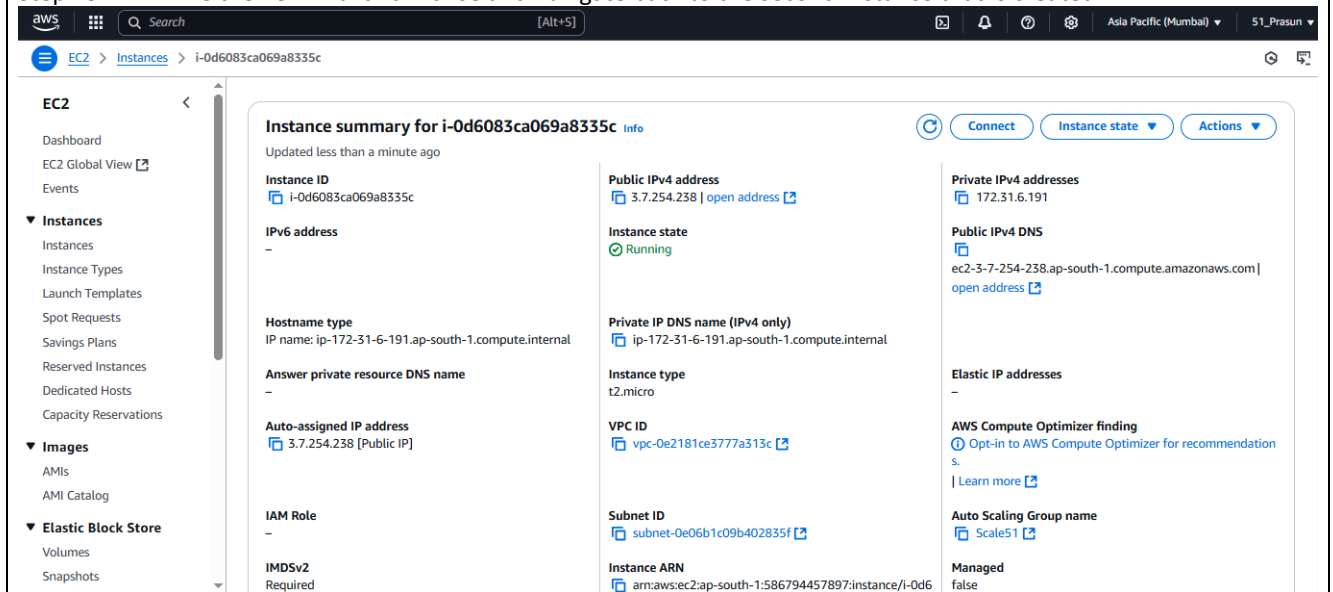
Step 23: Save the file, make it executable using the command “chmod 777 [filename]”.



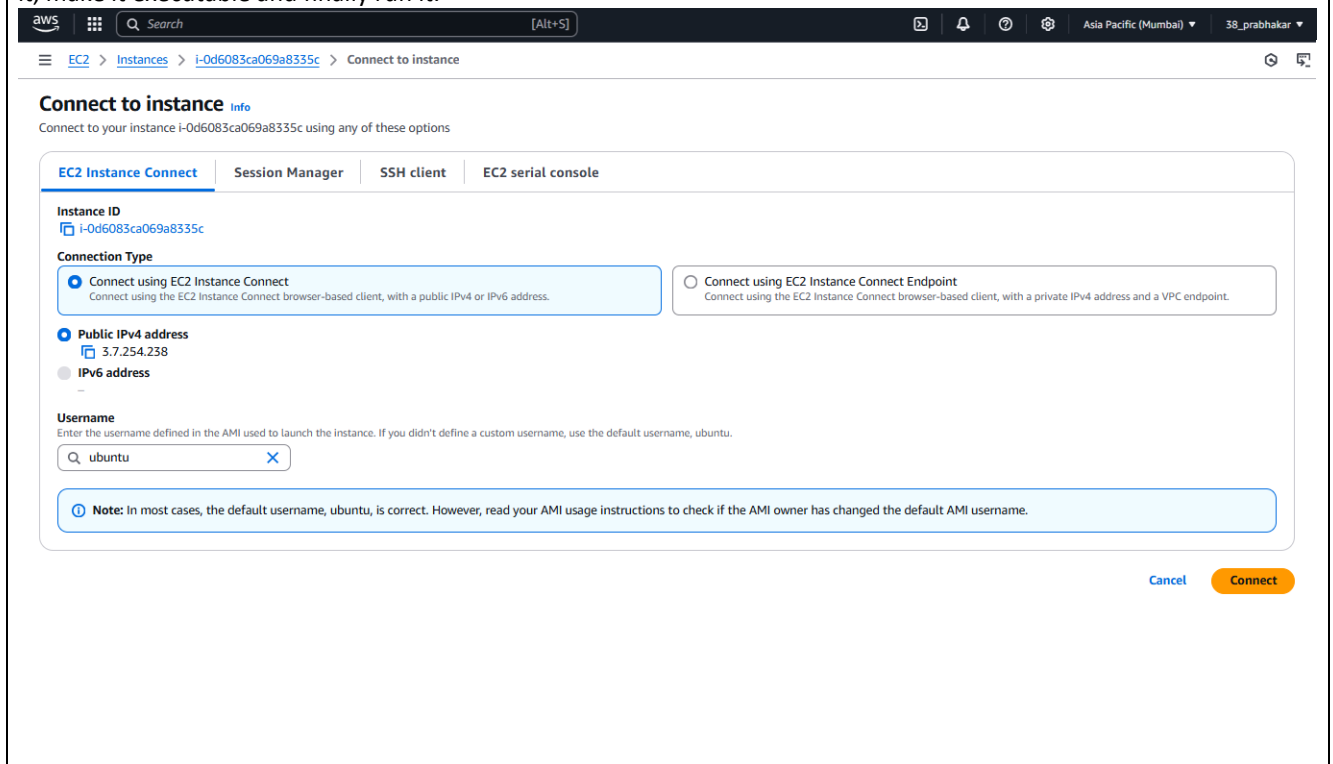
Step 24: Run the file using the command “./[filename]”



Step 25: Minimize the Terminal and Bitvise and navigate back to the second instance that is created.



Step 26: Click on Connect and follow the same steps as we've done in the Bitvise to create a new file, write the code in it, make it executable and finally run it.



A screenshot of a terminal window. The window has a dark background. The title bar at the top shows the AWS logo, a search icon, and the text "Search". On the right side of the title bar, there are icons for a terminal, a bell, a magnifying glass, and a refresh icon. The main area of the terminal is black with white text. The word "Looping" is printed repeatedly on each line, creating a vertical column of text on the left side of the terminal. The text is in a monospaced font. The window also shows a system tray at the bottom right with the text "Asia Pacific (Mumbai)" and "38_prabhakar".

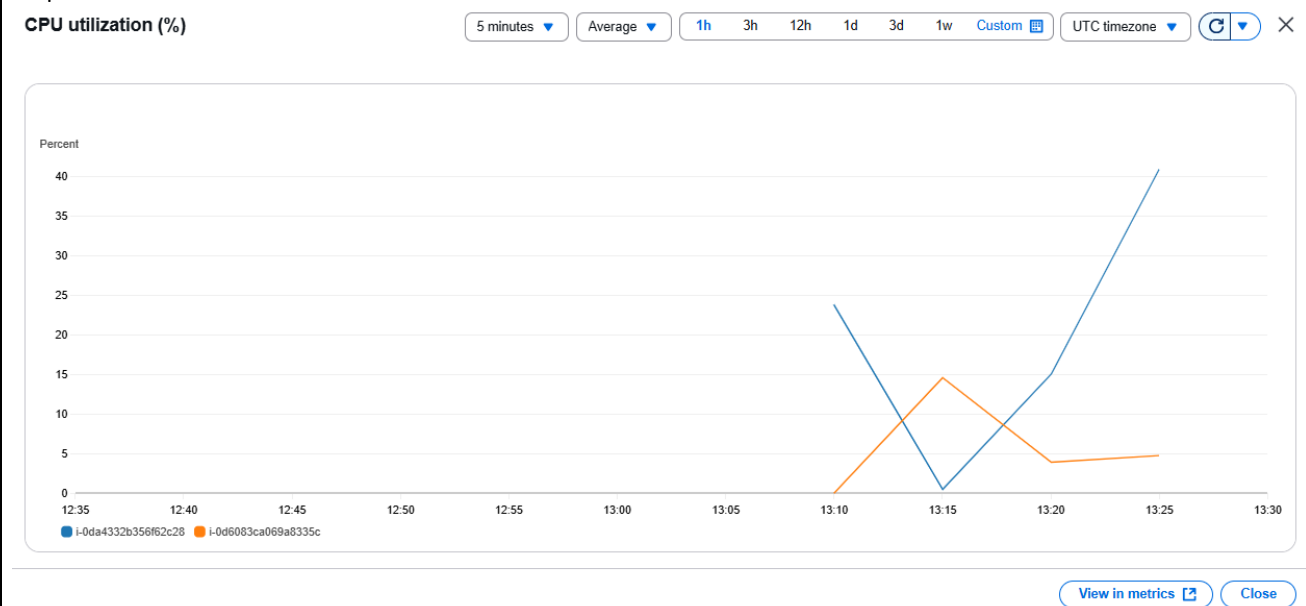
Step 27: Go to the previous tab and select both the instances to check the CPU Utilization.

The screenshot shows the AWS Management Console 'Instances' page. Two instances are selected, highlighted in blue. The table below summarizes their details:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
i-0da4332b356f62c28	i-0da4332b356f62c28	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-13-2
i-0d6083ca069a8335c	i-0d6083ca069a8335c	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-3-7-

Below the table, the '2 instances selected' section shows various metrics. The 'CPU utilization (%)' metric is highlighted, showing a line graph for the selected instances.

Step 29: Maximize the CPU Utilization window to have a better view.



So, finally we've Build scaling plans in AWS that balance the load on different EC2 instances.